

Title

Determinants and Predictive Modeling of Healthcare Expenditure

Abstract

This study examines the determinants of healthcare expenditure using statistical inference and predictive modeling techniques. Using a dataset of 2,000 individuals, we analyze the impact of demographic, behavioral, and clinical factors on annual medical costs. Smoking status emerges as the dominant cost driver, with an exceptionally large effect size (Cohen's $d = 2.48$). Linear regression achieves strong predictive performance ($R^2 \approx 0.61$), outperforming regularized and non-linear models. Cross-validation confirms model stability. The results suggest an additive cost structure dominated by lifestyle-related risk factors.

Introduction

Healthcare expenditure varies substantially across individuals due to differences in lifestyle, demographic characteristics, and underlying health conditions. Understanding the drivers of medical costs is crucial for healthcare planning and policy design. This study investigates which factors most strongly influence medical costs and evaluates the predictive performance of statistical and machine learning models.

Data Description

The dataset consists of 2,000 observations and 14 variables, including demographic information, lifestyle indicators, and clinical measures. The target variable is annual medical costs. No missing values or duplicate records were identified, ensuring data completeness.

Methodology

Exploratory Data Analysis

Exploratory analysis included distributional plots, correlation analysis, and grouped summaries. Medical costs exhibited right-skewness, and smoking prevalence was approximately 24%.

Statistical Inference

Group differences between smokers and non-smokers were assessed using Welch's t-test due to unequal variances (Levene's test $p < 0.001$). Effect size was quantified using Cohen's d. The Welch test confirmed a statistically significant difference in mean medical costs between the two groups:

- $t = -41.31$
- $p < 0.001$

Effect size was quantified using Cohen's d, yielding:

- Cohen's $d = 2.48$

This value indicates an exceptionally large practical effect, suggesting that smoking status has a substantial real-world impact on healthcare expenditure.

Predictive Modeling

Multiple models were evaluated, including Linear Regression, Ridge, Lasso, and Random Forest. Performance was assessed using R^2 , RMSE, and MAE. Five-fold cross-validation was employed to evaluate generalization.

Results

Descriptive Statistics

The dataset consists of 2,000 individuals with demographic, behavioral, and clinical characteristics. The average annual medical expenditure is approximately 2,937 units. Smokers represent 24% of the sample ($n = 476$), while non-smokers account for 76% ($n = 1,524$). Medical costs exhibit noticeable right-skewness, indicating the presence of high-cost individuals.

Predictive Model Performance

Model	R^2	RMSE	MAE
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Linear Regression	0.615	1,645	2,742
RidgeCV	0.615	1,645	1,299
LassoCV	0.612	1,651	1,312
Random Forest	0.560	1,759	1,404

Linear regression achieved the highest explanatory power ($R^2 = 0.615$) and lowest prediction error. Regularization techniques did not significantly improve performance, indicating limited multicollinearity and an adequately specified linear structure. Random Forest underperformed relative to linear models, suggesting that the data-generating process is primarily additive and approximately linear rather than strongly non-linear.

Cross-Validation

Five-fold cross-validation confirmed stable generalization:

- Mean $R^2 = 0.598 \pm 0.033$
- Mean RMSE = $1,683 \pm 65$
- Mean MAE = $1,350 \pm 50$

The relatively small variability across folds indicates consistent model performance and limited overfitting.

Feature Interpretation

Regression coefficient analysis revealed:

Smoking status increases expected annual medical cost by approximately 4,523 units.

Each additional chronic condition increases cost by approximately 597 units.

Physical activity reduces expected cost by approximately 74 units per additional hour.

Other demographic and clinical variables exhibited comparatively smaller effects.

Smoking status is clearly the dominant determinant of medical expenditure.

Model Diagnostics

Residual analysis revealed mild heteroscedasticity, indicating that prediction error increases slightly with higher predicted costs. However, no severe structural misspecification was observed. A log transformation of the target variable slightly reduced heteroscedasticity but did not improve predictive performance.

Overall, the linear model remains appropriate and interpretable.

Discussion

The findings demonstrate that lifestyle-related risk factors, particularly smoking behavior, are the primary drivers of healthcare expenditure in this dataset. The exceptionally large effect size suggests that smoking status alone explains a substantial portion of variation in medical costs.

The superior performance of linear regression compared to Random Forest suggests that the cost structure is primarily additive. Regularization techniques did not meaningfully improve performance, reinforcing the adequacy of the baseline linear model.

These results highlight the importance of preventive healthcare strategies and behavioral interventions in cost management.

Limitations

This study relies on observational synthetic data, limiting causal interpretation. Interaction effects and temporal dynamics were not explicitly modeled. Prediction intervals were not derived, and uncertainty quantification could be further expanded. Additionally, the dataset does not include external socioeconomic or healthcare access variables that may influence expenditure.

Conclusion

Smoking status is the dominant determinant of healthcare expenditure in this dataset, exerting a substantial and statistically significant impact. Linear regression provides strong predictive performance and stable generalization across cross-validation folds. The analysis suggests that healthcare costs follow a primarily additive and approximately linear structure dominated by lifestyle and clinical risk factors.

This study demonstrates a complete workflow combining statistical inference, model comparison, diagnostics, and interpretation, providing a robust framework for healthcare cost modeling.